

```

import pandas as pd
import numpy as np

df = pd.read_excel("Lab Session Data.xlsx", sheet_name="Purchase data")

X = df[['Candies (#)', 'Mangoes (Kg)', 'Milk Packets (#)']].values
y = df[['Payment (Rs)']].values

print("Dimensionality:", X.shape[1])
print("Number of Vectors:", X.shape[0])

rank = np.linalg.matrix_rank(X)
print("Rank:", rank)

cost = np.linalg.pinv(X) @ y
print("Product Costs:")
print(cost)

```

Output: Dimensionality: 3 Number of Vectors: 10 Rank: 3 Product Costs: [[1.] [55.] [18.]]

```

import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity

df = pd.read_excel("Lab Session Data.xlsx", sheet_name="thyroid0387_UCI")

df = pd.get_dummies(df)

v1 = df.iloc[0].values.reshape(1,-1)
v2 = df.iloc[1].values.reshape(1,-1)

cos = cosine_similarity(v1, v2)

print("Cosine Similarity:", cos[0][0])

```

Output: Cosine Similarity: 0.9999999999999999

```

import pandas as pd
from sklearn.preprocessing import MinMaxScaler

df = pd.read_excel("Lab Session Data.xlsx", sheet_name="thyroid0387_UCI")

numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns

scaler = MinMaxScaler()

df[numeric_cols] = scaler.fit_transform(df[numeric_cols])

print(df.head())

```

```

***      Record ID      age sex on thyroxine query on thyroxine \
0  0.000000e+00  0.000427  F      f      f
1  3.410871e-08  0.000427  F      f      f
2  9.891527e-07  0.000610  F      f      f
3  6.934301e-05  0.000534  F      f      f
4  6.937712e-05  0.000473  F      f      f

      on antithyroid medication sick pregnant thyroid surgery I131 treatment ... \
0      f      f      f      f      f      f ...
1      f      f      f      f      f      f ...
2      f      f      f      f      f      f ...
3      f      f      f      f      f      f ...
4      f      f      f      f      f      f ...

      TT4 measured  TT4  T4U measured  T4U  FTI measured  FTI  TBG measured  TBG  \
0      f      ?      f      ?      f      ?      f      ?
1      t  128      f      ?      f      ?      f      ?
2      f      ?      f      ?      f      ?      t  11
3      f      ?      f      ?      f      ?      t  26
4      f      ?      f      ?      f      ?      t  36

      referral source      Condition
0      other  NO CONDITION
1      other  NO CONDITION
2      other  NO CONDITION
3      other  NO CONDITION
4      other      S

```

Output: [5 rows x 31 columns]

```

import pandas as pd

df = pd.read_excel("Lab Session Data.xlsx", sheet_name="thyroid0387_UCI")

print(df.dtypes)
print(df.describe())
print(df.isnull().sum())

print(df.mean(numeric_only=True))
print(df.var(numeric_only=True))

```

Record ID	int64
age	int64
sex	object
on thyroxine	object
query on thyroxine	object
on antithyroid medication	object
sick	object
pregnant	object
thyroid surgery	object
I131 treatment	object
query hypothyroid	object
query hyperthyroid	object
lithium	object

goitre	object
tumor	object
hypopituitary	object
psych	object
TSH measured	object
TSH	object
T3 measured	object
T3	object
TT4 measured	object
TT4	object
T4U measured	object
T4U	object
FTI measured	object
FTI	object
TBG measured	object
TBG	object
referral source	object
Condition	object
dtype:	object

	Record ID	age
count	9.172000e+03	9172.000000
mean	8.529473e+08	73.555822
std	7.581969e+06	1183.976718
min	8.408010e+08	1.000000
25%	8.504090e+08	37.000000
50%	8.510040e+08	55.000000
75%	8.607110e+08	68.000000
max	8.701190e+08	65526.000000

Record ID	0
age	0
sex	0
on thyroxine	0
query on thyroxine	0
on antithyroid medication	0
sick	0
pregnant	0
thyroid surgery	0
I131 treatment	0
query hypothyroid	0
query hyperthyroid	0
lithium	0
goitre	0

tumor	0
hypopituitary	0
psych	0
TSH measured	0
TSH	0
T3 measured	0
T3	0
TT4 measured	0
TT4	0
T4U measured	0
T4U	0
FTI measured	0
FTI	0
TBG measured	0
TBG	0
referral source	0
Condition	0
dtype: int64	
Record ID	8.529473e+08
age	7.355582e+01
dtype: float64	
Record ID	5.748625e+13
age	1.401801e+06
dtype: float64	